- Midterm 1 is in 2 weeks. It will be in-class, and based on Homeworks 1-4.

<u>Prop</u> (Archimedean property) $\forall x \in \mathbb{R}$, $\exists n \in \mathbb{N}$ s.t. $n > x$.

<u>Pf</u>: Suppose not: $\exists x_0 \in \mathbb{R}$ s.t. $\forall n \in \mathbb{N}$, $n \leq x_0$.
Then $\mathbb{N}$ would satisfy:

- $\mathbb{N} \neq \emptyset$: since $1 \in \mathbb{N}$
- $\mathbb{N}$ is bounded above: by x_0

So, by the least upper bound property, $\exists L = \sup \mathbb{N} \in \mathbb{R}$.
As $L-1 < L$, then $L-1$ cannot be an upper bound for $\mathbb{N}$, so $\exists n_0 \in \mathbb{N}$ s.t. $n_0 > L-1$. But then $n_0 + 1 \in \mathbb{N}$ and $n_0 + 1 > L$, so L is not an upper bound — a contradiction. $\square$

<u>Cor</u> $\forall \varepsilon > 0$, $\exists n \in \mathbb{N}$ s.t. $\frac{1}{n} < \varepsilon$.

<u>Pf</u>: Apply the Archimedean property to $x = \frac{1}{\varepsilon}$. $\square$

<u>lem</u> $\forall a \in \mathbb{R}$, $\exists N \in \mathbb{Z}$ s.t. $N \leq a < N+1$.

<u>Pf</u>: Case: $a = 0$. Then $N = 0$ works.

Case: $a > 0$. Consider $A = \{n \in \mathbb{Z} : n \leq a\}$.

Then:

- $A \neq \emptyset$: $0 \in A$
- A is bounded above: by a

So, by the least upper bound property, $\exists L \in \mathbb{R}$

with $L = \sup A$. As $L-1 < L$, then $L-1$ cannot be an upper bound for A, so $\exists N \in A$ s.t. $N > L-1$. Altogether, we get

$$\left.\begin{array}{l} N \in A \Rightarrow N \leq a \\ N+1 > L \Rightarrow N+1 \notin A \Rightarrow N+1 > a \end{array}\right\} \Rightarrow N \leq a < N+1$$

as desired.

Case: $a < 0$. Exc. $\square$

<u>Prop</u> ($\mathbb{Q}$ is dense in $\mathbb{R}$) If $x, y \in \mathbb{R}$ satisfy $x < y$, then $\exists q \in \mathbb{Q}$ s.t. $x < q < y$.

<u>Pf</u>: Fix $x, y \in \mathbb{R}$ with $x < y$. Then $y - x > 0$, so $\exists n \in \mathbb{N}$ s.t.

$$\frac{1}{n} \overset{①}{<} y - x.$$

On the other hand, applying the lemma to $nx \in \mathbb{R}$, $\exists m \in \mathbb{Z}$ s.t.

$$m \overset{②}{\leq} nx \overset{③}{<} m+1.$$

Together, we conclude

$$x \underset{③}{<} \frac{m+1}{n} = \frac{m}{n} + \frac{1}{n} \underset{②}{\leq} x + \frac{1}{n} \underset{①}{<} y.$$

So $q = \frac{m+1}{n} \in \mathbb{Q}$ satisfies $x < q < y$. $\square$

<u>Cor</u> ($\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$) If $x, y \in \mathbb{R}$ satisfy $x < y$, then $\exists r \in \mathbb{R} \setminus \mathbb{Q}$ s.t. $x < r < y$.

<u>Pf</u>: As $x + \sqrt{2} < y + \sqrt{2}$, by the previous Proposition $\exists q \in \mathbb{Q}$ s.t.

$$x+\sqrt{2} < q < y+\sqrt{2} \quad \Rightarrow \quad x < q-\sqrt{2} < y.$$

So $r = q-\sqrt{2}$ works. Note that $r \notin \mathbb{Q}$; otherwise, $r \in \mathbb{Q} \Rightarrow \sqrt{2} = q - r \in \mathbb{Q}$, a contadiction. $\square$

	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	$\mathbb{C}$
order relation?	✓	✓	✓	✓	✓
field?	×	×	✓	✓	✓
ordered field?			✓	✓	×
LUB property?			×	✓	
Countable?	✓	✓	✓	×	

HW3

Next

<u>Prop</u> The set $A = \{ f : \mathbb{N} \to \{0,1\} \text{ is a function} \}$ is uncountable.

<u>Pf</u>: Suppose towards a contradiction that A were countable. Then we can enumerate the elements of A:

$$\begin{array}{lllll}
f_1: & f_1(1) & f_1(2) & f_1(3) & \cdots \\
f_2: & f_2(1) & f_2(2) & f_2(3) & \cdots \\
f_3: & f_3(1) & f_3(2) & f_3(3) & \cdots \\
\vdots & & & &
\end{array}$$

Define $g: \mathbb{N} \to \{0,1\}$ by $g(n) = 1 - f_n(n)$, so that the nth value is the opposite of the nth value of f_n. Then $g \in A$, but $g \neq f_n \ \forall n$ since $g(n) \neq f_n(n)$. Therefore we have not

enumerated all the elements of A — a contradiction. $\square$

<u>Rmk</u> This trick is called "Cantor's diagonal argument."

<u>Cor</u> The set $[0,1) = \{x \in \mathbb{R} : 0 \le x < 1\}$ is uncountable.

<u>Idea:</u> Binary expansion

$$x \in [0,1) \implies x = 0.x_1 x_2 x_3 \cdots, \quad x_n \in \{0,1\}.$$

- We require that no expansion ends in all 1s: e.g. $0.0111\ldots \rightsquigarrow 0.1000\ldots$
- Same argument works

SEQUENCES (Ross Ch. 2, Rudin Ch. 3)

<u>Def</u> A <u>sequence</u> of real numbers is a function $f : \mathbb{N} \to \mathbb{R}$. We denote the sequence by $f(1), f(2), f(3), \ldots$, or $\{f(n)\}_{n \ge 1}$, or $\{f_n\}_{n \ge 1}$.

<u>Ex</u> ① $\{a_n\}_{n \ge 1}$ with $a_n = \frac{1}{n}$ is the sequence $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$

② $\{a_n\}_{n \ge 1}$ with $a_n = n^2$ is the sequence $1, 4, 9, 16, \ldots$

③ $\{a_n\}_{n \ge 1}$ with $a_n = (-1)^n$ is the sequence $-1, 1, -1, 1, \ldots$. Note that the sequence $\{(-1)^n\}_{n \ge 1}$

is not the same thing as the set of values $\{-1, 1\}$.